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Size, Returns, and Value: Do Private Equity Firms Allocate Capital According to Manager Skill?

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1 Big picture

- **Question.** As private equity (PE) funds become larger, what is the right way to measure performance, is there persistent skill at the individual-manager level, and do PE firms allocate more capital to the managers who create more value?
- **Setting.** The paper studies buyout PE using a proprietary deal-level cash-flow data set linked, whenever possible, to the individual managers leading each deal. The emphasis is on how scale changes the meaning of performance persistence.
- **Core challenge.** Relative returns (for example, PME) can decline as funds and deals scale up, yet managers may still create more value in dollar terms. A performance metric based only on relative returns can therefore miss economically important skill.
- **Main conceptual move.** The paper argues that **net present value (NPV)** is the right measure of PE manager skill when scale matters. NPV combines both returns and deal size, so it is the PE analogue of Berk and van Binsbergen’s value-added measure for mutual funds.
- **Main empirical result 1.** At the *fund* level, relative performance persistence measured by PME weakens over time, but NPV persistence strengthens. Larger funds earn lower relative returns, yet often higher dollar value-added.
- **Main empirical result 2.** At the *manager-deal* level, early in the sample there is persistence in PME, but in later years that persistence largely disappears. In contrast, persistence in NPV becomes stronger.
- **Main empirical result 3.** PE firms appear to use internal labor markets to match capital with skill. Managers with stronger NPV track records are assigned larger subsequent deals, are more likely to be promoted to partner, and are less likely to leave the PE industry.
- **Main organizational result.** About **40% of NPV** is attributed to the PE firm’s internal capital allocation decisions. This is the paper’s strongest statement about the role of the organization itself rather than the individual deal maker.
- **Why this matters.** The paper changes the way one should think about performance persistence in PE. Declining persistence in relative returns does *not* imply disappearing skill; it may instead reflect the fact that successful managers are asked to deploy more capital.
- **Economic takeaway.** PE firms create value not just by selecting deals, but by reallocating scarce internal capital toward the managers who have shown the strongest ability to generate NPV. The organizational problem is therefore a *matching-capital-to-skill* problem.

2 Data and measurement (key objects)

2.1 Data sources and sample construction

- **Core PE cash-flow data.** The paper uses a proprietary buyout deal-level database assembled from three large institutional investors. For nearly all PE firms, the authors observe the complete investment history across multiple funds in the same fund family.
- **What is observed.** For each deal, the data include monthly gross cash flows into and out of the portfolio company, along with the latest fair value when the deal is not yet fully realized.

- **Manager attribution.** The authors hand-assemble manager-deal matches from Preqin, PitchBook, MergerMarket, VentureSource, and the *Who's Who in Private Equity* volumes. Board seats and reported lead-deal involvement are used to identify which individual managers were responsible for a transaction.
- **Career data.** To study promotion and exit, the authors collect manager job titles, start dates, and end dates from BoardEx, LinkedIn, PitchBook, VentureSource, and manual web searches.
- **Fund economics.** For a subsample of 328 funds, they also obtain private placement memoranda and limited partnership agreements to measure management fees, carried interest, and hurdle rates.

2.2 Sample size and coverage

- **Funds.** 940 buyout funds raised by 309 PE firms, with vintage years from 1974 to 2011.
- **Deals.** 12,910 unique deals, with investments occurring from 1974 to 2015 and cash-flow / fair-value observations extending to the end of 2018.
- **Managers.** 3,857 unique managers are identified.
- **Matched sample.** Manager attribution is available for 4,873 deals, around 40% of the universe.
- **Manager-deal involvements.** Because many deals have multiple named managers, the matched sample yields 10,065 manager-deal involvements.

Interpretation. The paper's comparative advantage is not just that it has PE cash flows. It is that it links those cash flows to *individual managers*, making it possible to separate firm-level persistence from person-level skill.

2.3 Key performance measures

- **MOIC.** Multiple on invested capital, a widely used absolute-return measure.
- **PME.** Gross Kaplan–Schoar public market equivalent, calculated relative to regional MSCI indices (Asia, Europe, North America).
- **NPV.** Net present value of deal cash flows discounted using the same regional MSCI indices plus a premium of 200 basis points per year.

At the deal level, for manager i and transaction n ,

$$NPV_{i,n} = E \left[\sum_{t=0}^T (c_{n,t}^- + c_{n,t}^+) D_t^0 \right], \quad (1)$$

where $c_{n,t}^-$ denotes capital invested, $c_{n,t}^+$ denotes distributions received, and D_t^0 is the discount factor. At the fund level,

$$NPV_f = \sum_{n \in f} NPV_n. \quad (2)$$

Interpretation.

- PME is a *relative-return* concept.
- NPV is a *scale-adjusted value-creation* concept.
- The paper’s central claim is that when deal sizes become much larger, the second concept is the economically relevant one.

2.4 Why NPV is especially attractive in PE

- PE deals are “buy-to-sell” investments with finite lives, so realized deal-level cash flows can be observed directly.
- The paper uses latest fair values for unrealized deals and relies on earlier evidence showing that such valuations are, on average, unbiased predictors of future PE cash flows.
- Because the discount rate is based on realized public-market returns plus a PE premium, the authors interpret NPV as a kind of **scale-adjusted PME**.

2.5 Descriptive facts: fund level

- Mean fund size is \$931.2m and median fund size is \$330.9m.
- Mean gross MOIC is 2.44 and median gross MOIC is 2.04.
- Mean gross PME is 1.91 and median gross PME is 1.67.
- Mean fund NPV is \$333.8m and median fund NPV is \$103.5m.

Over time, fund sizes rise sharply while average relative returns fall. The mean fund size increases from \$380.5m in 1974–1989 to \$1,762.9m in 2005–2010, while mean PME falls from 2.22 to 1.55. Yet mean fund NPV generally rises over most of the sample before dipping in the final subperiod.

Interpretation. This already foreshadows the paper’s core point: returns and value creation do not move one-for-one when funds scale.

2.6 Descriptive facts: deal level

- Mean deal size is \$48.0m and median deal size is \$15.4m.
- Mean deal PME is 1.99 and median deal PME is 1.25.
- Mean deal NPV is \$24.3m and median deal NPV is only \$1.0m.
- Using the baseline discount rate, roughly **44% of deals have negative NPV ex post**.

Over time, mean deal size rises from \$21.5m before 1990 to \$77.6m in 2005–2010, while mean deal PME becomes more cyclical and somewhat weaker. Mean deal NPV rises strongly through 2000–2004 and then softens in the final period.

Interpretation. PE returns are strongly right-skewed, especially at the deal level. The median deal barely creates value, while the right tail drives much of aggregate NPV.

2.7 Backfill and survivorship concerns

- A serious concern in manager attribution data is that successful managers may be more likely to claim good deals and distance themselves from bad deals.
- The paper addresses this by triangulating several sources and by using the *Who's Who* directories, where deal involvement is often reported during the holding period, before the final outcome is known.
- The authors also compare their data against Burgiss-MSCI fund-level and deal-level data and find broadly similar return distributions.

Interpretation. The manager-level analysis is unusually data-intensive for PE, but it is also the part of the paper where measurement error is potentially most important. The authors explicitly confront this issue rather than ignoring it.

3 Models and empirical specifications (all equations + interpretation)

3.1 Incentives in private equity and why scale matters

The paper begins with the Berk–Green idea that success attracts more capital, and therefore relative returns may fall even when skill is present. In PE the main adjustment margin is *fund size*, not fees. Headline carried interest is usually around 20%, and management fees change little across successive funds. Hence successful PE firms have powerful incentives to raise larger funds and execute larger deals.

A useful benchmark from the mutual-fund literature is Berk and van Binsbergen’s value-added concept:

$$VA_f = q_{f,t-1} \alpha_f^g(q_{f,t-1}), \quad (3)$$

where $q_{f,t-1}$ is lagged AUM and α_f^g is gross alpha.

Interpretation. The paper’s NPV measure is the PE counterpart to this object. It explicitly rewards a manager who creates more dollars of value, even if relative returns decline as capital scales.

3.2 Deal-level performance and discounting

The baseline discount rate is constructed from out-turn public equity returns plus 200 bps, motivated by higher leverage and illiquidity in PE. The authors also discuss alternative estimates of PE beta and show that the results are robust to other discount-rate assumptions.

Interpretation. The point is not that 200 bps is the one true premium. The point is that NPV comparisons are made *within comparable times and places*, so moderate discount-rate variation does not overturn the ranking of managers or the persistence patterns.

3.3 Fund-level persistence regressions

At the fund level, the paper estimates regressions of the form

$$\begin{aligned} Perf_{f,t} = & \alpha + \rho Perf_{f,t-1} + \beta \log(FundSize_{f,t}) \\ & + \gamma \log(FundSequence_f) + \delta RealizationShare_{f,t} + FE + \varepsilon_{f,t}. \end{aligned} \quad (4)$$

where $Perf$ is either gross PME or fund NPV.

Interpretation.

- ρ measures persistence across successive funds of the same PE organization.
- The fund-size coefficient captures diseconomies of scale in *relative returns*.
- The contrast between the PME regressions and the NPV regressions is one of the paper’s central empirical results.

3.4 Manager-level persistence regressions

At the manager-deal level, the paper estimates equations of the form

$$\begin{aligned} Perf_{i,n} = & \alpha + \rho Perf_{i,n-1} + \beta \log(InvSize_{i,n}) \\ & + \gamma \log(HoldingPeriod_{i,n}) + \delta \log(DealSequence_{i,n}) \\ & + \theta RealizationStatus_{i,n} + FE + u_{i,n}. \end{aligned} \quad (5)$$

where $Perf$ is deal PME or deal NPV.

Interpretation.

- If PME persistence fades while NPV persistence strengthens, then the right reading is not “skill disappeared” but rather “skill became more visible in dollar value-added than in relative returns.”
- The negative coefficient on investment size in the PME regressions is the reduced-form sign of diseconomies of scale.

3.5 Skill and scalability decomposition

To separate skill from scalability, the authors adapt Barras, Gagliardini, and Scaillet (2022). For each manager they estimate

$$PME_{i,n} = a_i + b_i \log(InvSize_{i,n}) + c_i \log(DealSequence_{i,n}) + u_{i,n}. \quad (6)$$

- a_i is the manager’s **skill coefficient**: the gross PME the manager would deliver on a hypothetical \$1m deal.
- b_i is the manager’s **scale coefficient**: how much relative return declines as deal size increases.

The paper then estimates the distribution of these coefficients nonparametrically, with an errors-in-variables correction, using the sample of managers with at least five realized deals.

Interpretation. This is not a structural estimate of managerial ability. It is a disciplined way of asking whether most managers look value-creating at small scale and whether most of them face diseconomies of scale when trusted with larger deals.

3.6 Track-record measure for internal capital markets

To study whether PE firms allocate capital according to skill, the paper constructs an evolving track-record proxy. For manager i at time t ,

$$SuccessRate_{i,t} = \frac{1}{N_{i,t-1}} \sum_{n < t} \mathbf{1}\{NPV_{i,n} > Median(NPV \text{ of same PE-firm-year})\}. \quad (7)$$

Interpretation.

- A manager is labeled an *outperformer* if at least 50% of past realized deals beat the median deal NPV of the same PE firm in the same year.
- This is a relative measure inside the organization, so it is intended to capture the manager’s standing in the internal labor market.

The paper then estimates multivariate regressions of the form

$$Y_{i,t+1} = \alpha + \beta NPV SuccessRate_{i,t} + \Gamma X_{i,t} + FE + \varepsilon_{i,t}, \quad (8)$$

where Y is next deal size per manager, partner promotion, leaving the firm, or leaving the PE industry, and X includes age, gender, MBA, Ivy-League attendance, and deal sequence.

3.7 Capital-allocation contribution to NPV

The paper’s final estimating equation is

$$\begin{aligned} NPV_{i,n,t} = & \lambda_t + \lambda_r + \lambda_i + \lambda_b + \lambda_f \\ & + \beta \mathbf{1}\{\Delta InvSize/Mgr \text{ increases a lot}\} \\ & + \gamma \mathbf{1}\{\Delta InvSize/Mgr \text{ decreases a lot}\} \\ & + \bar{\delta} Controls + \epsilon_{i,n,t}. \end{aligned} \quad (9)$$

Here the omitted category is a deal whose investment size per manager is roughly unchanged relative to the manager’s previous deal.

Interpretation.

- If the coefficient on the “increase” dummy is positive, the paper interprets that as evidence that the PE firm’s capital-allocation decision itself contributes to value creation.
- The inclusion of manager, fund, year, region, and industry fixed effects tries to strip out many confounding sources of heterogeneity.
- This is the paper’s empirical vehicle for quantifying the organizational contribution to NPV.

4 Identification and credibility (what makes the estimates credible)

4.1 What identifies the central empirical claims

- **Fund-level persistence.** Identified from variation across successive funds within fund families, controlling for size, sequence, and realization.

- **Manager-level persistence.** Identified from the relationship between a manager’s current deal and prior deal outcomes, with controls for deal size, holding period, and fixed effects.
- **Skill versus scale.** Identified from within-manager variation in realized deal PME as deal sizes change.
- **Internal capital allocation.** Identified from the link between a manager’s NPV-based track record and the size of the next deal entrusted to that manager.
- **Organizational value creation.** Identified from whether large increases in deal size per manager predict larger NPV even after manager, fund, year, region, and industry fixed effects are included.

4.2 Why NPV is the right organizing variable

- The paper is not claiming PME is uninformative.
- It is claiming PME becomes a poor *summary statistic of skill* once managers are deliberately scaled up.
- A successful manager may earn a lower PME on a much larger deal while still adding more total value to investors and to the PE firm.

Interpretation. This is analogous to the Berk–van Binsbergen critique of net alpha in mutual funds: an equilibrium response to skill can compress relative returns without eliminating skill.

4.3 What makes the manager-attribution evidence believable

- The authors combine multiple public and proprietary sources rather than relying on one attribution database.
- They explicitly discuss backfill and survivorship bias.
- They exploit the fact that the *Who’s Who* series records deal involvement while deals are still alive, not just after successful exits.
- They compare their sample to Burgiss-MSCI and find broadly similar gross-return distributions.

4.4 What the fixed effects are doing

- **Year fixed effects** absorb market-wide conditions and deal-vintage patterns.
- **Region and industry fixed effects** absorb broad differences in opportunity sets and risk.
- **Manager fixed effects** absorb time-invariant manager ability in the capital-allocation-to-NPV regressions.
- **Fund fixed effects** absorb cross-fund heterogeneity in economics and strategy.

Interpretation. The regressions are not fully causal experiments, but the fixed effects make the paper’s comparisons much sharper than raw cross sections.

4.5 Main limitations

- Manager attribution is incomplete and noisy: only about 40% of deals are matched to managers.
- Track records in PE are short. Many managers have very few realized deals, which makes skill estimation noisy.
- Risk adjustment remains imperfect. Regional MSCI indices control for geography, but not fine industry differences or project-specific risk.
- The paper documents that better managers get more capital, but it cannot fully separate deliberate reallocation from the fact that stronger managers may source better or larger opportunities.
- The “size decrease” estimates are likely biased downward because the sample is right-censored: after a manager is sidelined or leaves, future deals may simply not be observed.

4.6 Relation to the literature

- To the PE persistence literature, the paper adds the idea that persistence should be studied in both relative-return and dollar-value terms.
- To the mutual-fund literature, it imports the distinction between skill and scale and applies it to closed-end PE funds.
- To the literature on organizations and internal labor markets, it shows that PE firms appear to use both *capital allocation* and *career progression* to reward skill.
- To the “individual versus organization” literature, it shows that both matter: managers have skill, but the PE organization creates additional value by matching larger deals to those skilled managers.

5 Tables and figures

5.1 Table I and Table II: what the raw data look like

Read:

- Table I shows that fund size rises enormously over time, while gross PME falls and NPV generally rises.
- The striking comparison is between the 2000–2004 and 2005–2010 vintages: mean fund size roughly doubles, PME falls sharply, and mean NPV slips from \$467.8m to \$330.3m.
- Table II shows that deal-level returns are even more skewed than fund-level returns. Median deal PME is only 1.25, and median NPV is just \$1.0m.
- The matched manager sample consists of larger, somewhat more recent deals, but average returns remain similar to the full sample. That is reassuring for external validity.

Table I
Fund Characteristics

This table presents univariate statistics for a sample of 940 buyout funds. It reports the mean and median values for the following variables: fund size (in millions of 2000 U.S. dollars), the multiple on invested capital (MOIC) gross return, the gross Kaplan-Schoar public market equivalent (PME) gross return (calculated relative to three regional MSCI indices: Asia, Europe, and North America, in local currency), and fund net present value (NPV). Fund NPV is the NPV of all deals in a fund, discounted using the same three regional MSCI indices plus 200 bps p.a. Fund vintage years are based on the year in which the first investment was made.

	Obs.	Fund Size		MOIC		PME		NPV	
		Mean	Median	Mean	Median	Mean	Median	Mean	Median
Full sample	940	931.2	330.9	2.44	2.04	1.91	1.67	333.8	103.5
<i>Regions</i>									
North America	458	1,218.0	487.5	2.56	2.06	1.88	1.69	389.4	126.5
Europe	426	687.7	238.5	2.34	2.04	1.91	1.66	302.3	91.9
RoW	56	438.1	262.9	2.26	1.90	2.07	1.63	118.0	52.5
<i>Fund Vintage Years</i>									
1974–1989	75	380.5	138.6	4.31	3.23	2.22	1.74	68.8	36.0
1990–1994	112	359.7	173.1	3.22	2.77	1.96	1.73	139.6	46.2
1995–1999	245	621.4	313.2	2.43	2.07	2.08	1.85	363.1	116.6
2000–2004	263	845.4	331.6	2.38	2.22	1.96	1.84	467.8	168.0
2005–2010	245	1,762.9	658.5	1.60	1.39	1.55	1.42	330.3	124.5

Table III
Scale and Performance by Fund Sequence

This table shows mean values for our sample of 940 buyout funds. Panel A tabulates gross fund performance by fund sequence. Relative returns are measured by the multiple on invested capital (MOIC) and the gross Kaplan-Schoar public market equivalent (PME) returns, calculated relative to three regional MSCI indices (Asia, Europe, and North America) in local currency. Fund NPV is the net present value of all deals in a fund, discounted using the same three regional MSCI indices plus 200 bps p.a. Panel B shows fund size, the number of investments per fund, and average investment size, again by fund sequence. Both size variables are in millions of 2000 U.S. dollars. For a subsample of 328 funds with available information on compensation, Panel C shows management fees (in % of AUM), carried interest (in % of profits), and the hurdle rate (in %). Panel D extrapolates the fee information in Panel C to all of our sample funds and calculates the corresponding (unweighted) total gain split between the fund's LPs and the PE firm (in millions of 2000 U.S. dollars). Note that fund size reflects committed capital, and is not the same as the number of deals multiplied by average investment size (as not all funds have fully invested and some portion of committed capital will be used for management fees).

	Fund Sequence				
	First <i>n</i> = 280	Second <i>n</i> = 229	Third <i>n</i> = 158	Fourth <i>n</i> = 104	Fifth and Higher <i>n</i> = 169
Panel A: Gross performance					
MOIC	2.44	3.07	2.42	2.13	2.09
PME	1.91	2.21	1.91	1.82	1.67
NPV	333.7	237.6	184.9	376.8	511.5
Panel B: Scale					
Fund Size	931.2	372.3	529.5	838.5	1,302.3
Number of Deals	15.2	14.5	12.8	14.9	17.2
Avg. Investment Size	44.9	21.1	35.2	42.4	66.3
Panel C: Fund economics					
Management Fee	1.83	2.00	1.96	1.81	1.78
Carried Interest	20.27	20.38	20.06	20.34	20.33
Hurdle Rate	7.95	8.05	8.18	7.80	7.83
Panel D: NPV split					
Limited Partners	212.5	154.1	124.6	228.4	319.1
PE Firm	121.4	83.6	60.5	148.4	192.6
PE Firm/100 Total NPV	36.4%	35.2%	32.7%	39.4%	37.6%

Figure 1. Fund-level gross PME and NPV. This figure shows Whisker box plots of absolute Kaplan-Schoar gross public market equivalents (PME) and net present value (NPV) for our sample of 940 buyout funds by annual fund size quartiles. PME is calculated relative to three regional MSCI indices (Asia, Europe, and North America) in local currency. Fund NPV is the NPV of all deals in a fund, discounted using the same three regional MSCI indices plus 200 bps p.a. The boxes represent the 25th, 50th (median), and 75th percentiles. As common with Whisker versions of box plots, the adjacent lines indicate values 1.5 times the interquartile range above (below) the 75th (25th) percentile. (Color figure can be viewed at wileyonlinelibrary.com)

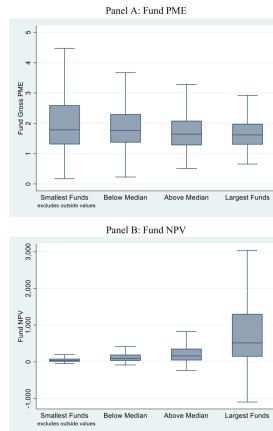


Table II
Deal Characteristics

Panel A shows univariate statistics at the deal level for three different samples. The overall sample of 940 funds contains 12,910 unique deals (*full sample*). For 4,873 of these deals, we could identify at least one individual manager running the deal for a GP organization (*matched sample*). As some deals were managed by several individuals from one PE firm, or as club deals were managed by individual managers from different PE firms, we observe 10,065 *deal involvements* (i.e., manager-unique deal dyads). We report the multiple on invested capital (MOIC) gross return, the gross Kaplan-Schoar public market equivalent (PME) return, calculated relative to three regional MSCI indices (Asia, Europe, and North America) in local currency, and deal net present value (NPV), discounted using the same three regional MSCI indices plus 200 bps p.a. *Investment Size* denotes the PE firm's investment measured in millions of 2000 US dollars. *Holding Period* indicates the difference between an investment's date and the exit date or the latest reported valuation (in the case of unrealized transactions), measured in years. *Investment Year* is measured based on the first cash flow into a deal. *Realization Status* is a dummy variable set to one if a deal is at least partially realized, and zero otherwise. *Managers on Deal* indicates the number of managers we were able to assign to a given deal. In Panel B, we report equity investment size (in millions of 2000 U.S. dollars), PME, and NPV for several time categories.

Variables	Panel A: Deal characteristics by sample type					
	Full Sample (<i>n</i> = 12,910)		Matched Sample (<i>n</i> = 4,873)		Deal Involvements (<i>n</i> = 10,065)	
	Mean (1)	Median (2)	Mean (3)	Median (4)	Mean (5)	Median (6)
Investment Size	48.0	15.4	66.3	27.6	91.1	35.2
MOIC	2.62	1.53	2.67	1.56	2.47	1.52
PME	1.99	1.25	2.16	1.36	2.03	1.35
NPV	24.3	1.0	37.3	5.4	47.3	6.8
<i>thereof positive</i>	56%		61%		61%	
Holding Period	4.25	3.75	4.10	3.67	4.02	3.67
Investment Year	2001	2001	2003	2004	2004	2005
Realization Status	0.83	1.00	0.77	1.00	0.74	1.00
Managers on Deal	-	-	2.07	2.00	3.21	3.00

Panel B: Deal characteristics over time (full sample)

	Obs.	Investment Size		PME		NPV	
		Mean	Median	Mean	Median	Mean	Median
1974–1989	792	21.5	2.7	2.15	0.89	1.6	–0.3
1990–1994	1,349	16.1	5.5	1.81	1.23	6.6	0.1
1995–1999	3,087	35.2	10.7	2.20	1.25	17.5	0.6
2000–2004	3,565	42.7	16.8	2.13	1.50	34.6	3.7
2005–2010	4,117	77.6	31.6	1.75	1.15	30.7	2.7

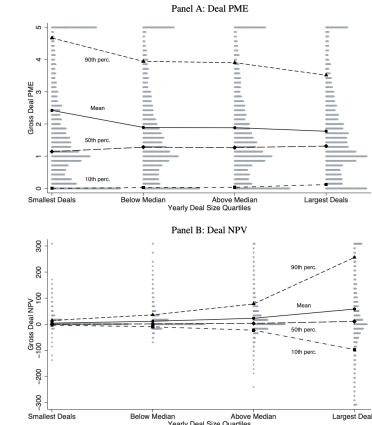


Figure 2. Deal-level gross PME and NPV. This figure shows Kaplan-Schoar gross public market equivalents (PME) and net present value (NPV) for our sample of 12,910 unique deals by annual investment size quartiles. Investment size quartiles are computed by investment year. But, in each deal in our sample is assigned to a quartile based on its equity investment size compared to all other deals from the same investment year. PME is calculated relative to three regional MSCI indices (Asia, Europe, and North America) in local currency. NPV is the NPV of the deal (in millions of 2000 U.S. dollars), discounted using the same three regional MSCI indices plus 200 bps p.a. Investment size is measured in millions of 2000 US dollars. Each panel displays unweighted average, as well as the 10th, 50th, and 90th percentiles by deal size quartile. For illustration purposes, performance values larger than a gross PME (NPV) of 5 (\$300m) are grouped into one bracket. (Color figure can be viewed at wileyonlinelibrary.com)

5.2 Table III and Figure 1 / Figure 2: scaling up lowers relative returns but not necessarily value

Read:

- Table III is the fund-level counterpart of the paper’s main argument.
- Across fund sequence, average gross PME falls from 2.21 for first funds to 1.62 for fifth-and-later funds, while average fund NPV rises from \$237.6m to \$543.8m.
- The same table shows that PE firms scale mainly by **larger deals**, not by many more deals: average investment size rises from \$21.1m for first funds to \$86.5m for fifth-and-later funds, while the number of deals moves only modestly.
- Panel C shows that fees and carry barely move with success, reinforcing the incentive to scale AUM rather than to charge much higher headline fees.
- Panel D is economically important: around **36%** of gross fund NPV accrues to the PE firm in fees and carry, with a fairly stable share across fund sequence.
- Figure 1 shows visually that larger funds have lower PMEs but much higher and more dispersed NPVs.
- Figure 2 shows the same logic at the deal level: larger deals have lower relative returns on average, but much wider NPV distributions and a higher right tail.

5.3 Table IV: fund persistence switches from PME to NPV

Read:

- In early years, prior-fund PME strongly predicts current-fund PME: the coefficient is 0.178, very close to Kaplan–Schoar’s classic result.
- In later years, PME persistence drops to 0.043 and loses significance.
- For NPV, the pattern flips. Prior-fund NPV predicts current-fund NPV in both subsamples, and the coefficient is larger in later years (0.243 versus 0.189).
- The negative size coefficient in the PME regression for later years tells you that scale mechanically erodes relative returns.

Bottom line. Looking only at PME would suggest that persistent skill is fading away. Looking at NPV says that persistent value creation is still very much present.

5.4 Table V: manager persistence

Read:

- At the manager-deal level, the same pattern appears even more cleanly.
- For realized deals up to 2004, the prior-deal PME coefficient is 0.123 and significant.
- For realized deals after 2004, the prior-deal PME coefficient falls to 0.055 and is no longer significant.

Table IV
Fund Persistence Regressions

This table reports results of OLS regressions of the investment performance of a PE firm's current fund on the performance of the firm's previous fund. In models (1) to (3), we look at the gross Kaplan-Schoar PME return, calculated relative to three regional MSCI indices (Asia, Europe, and North America) in local currency. In models (4) to (6), we examine fund NPV in millions of U.S. dollars, discounted using the same three regional MSCI indices plus 200 bps p.a. Early years are all funds closed prior to 1996, while later years range from 1996 to 2011 (see Kaplan and Schoar (2005)). *Log Fund Size* is the natural logarithm of the total committed capital in millions of 2000 US dollars. *Log Fund Sequence* is the natural logarithm of fund sequence within a PE firm's fund family. *Fund Realization Share* measures the share of deals in a fund that have been at least partially realized. This variable drops from models (2) and (4) because all funds are fully realized in the early years. All regressions include vintage year and fund region fixed effects. Standard errors are in parentheses and clustered at the PE firm level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Variables	Fund PME			Fund NPV		
	All	Early	Later	All	Early	Later
	(1)	(2)	(3)	(4)	(5)	(6)
Gross PME Previous Fund	0.058* (0.031)	0.178* (0.106)	0.043 (0.032)			
NPV Previous Fund				0.243*** (0.039)	0.189** (0.087)	0.243*** (0.042)
Log Fund Size	-0.082*** (0.029)	0.055 (0.097)	-0.100*** (0.031)			
Log Fund Sequence	-0.045 (0.082)	-0.070 (0.283)	-0.045 (0.085)	253.769*** (73.732)	-126.528 (143.499)	272.722*** (83.776)
Fund Realization Share	0.422*** (0.130)		0.445*** (0.133)	491.315*** (124.971)		542.317*** (135.834)
Observations	612	89	523	612	89	523
Adj. R^2	0.088	-0.002	0.103	0.134	0.038	0.127
Vintage Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes

Table V
Manager Persistence

This table reports results of OLS regressions of the gross public market equivalent (PME) and deal net present value (NPV) for each manager's current deal on the corresponding performance (PME or NPV, respectively) of the manager's previous deal. For the NPV, we use three regional MSCI indices plus 200 bps p.a. to discount deal gross cash flows. The analysis is at the manager-deal involvement level. In models (1) and (5), we use the entire sample of managers for which we observe at least two deals. In models (2) to (4) and (6) to (8), we restrict our sample to at least partially realized deals. Models (3) and (7) further restrict the sample to deals up until 2004 (the median investment year), while models (4) and (8) only include deals after 2004. All regressions include investment year, deal region, and deal industry fixed effects. Standard errors are in parentheses and clustered at the manager level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Dependent Variable	Deal PME				Deal NPV			
	Realized deals				Realized deals			
	All deals (1)	All (2)	Year ≤ 2004 (3)	Year > 2004 (4)	All deals (5)	All (6)	Year ≤ 2004 (7)	Year > 2004 (8)
Previous Deal Performance	0.087*** (0.025)	0.099*** (0.031)	0.123*** (0.041)	0.055 (0.050)	0.092*** (0.025)	0.172*** (0.045)	0.121*** (0.030)	0.196*** (0.066)
Log Investment Size	-0.242*** (0.047)	-0.286*** (0.068)	-0.325** (0.126)	-0.265*** (0.059)				
Log Holding Period	0.125*** (0.035)	0.176*** (0.066)	-0.198* (0.120)	0.558*** (0.067)	4.680 (3.883)	29.614*** (6.165)	-15.837* (8.816)	72.029*** (9.625)
Log Manager Deal Sequence	-0.029 (0.050)	-0.078 (0.067)	-0.145 (0.126)	-0.059 (0.088)	-2.740 (6.057)	-2.551 (8.229)	-5.754 (7.481)	-14.070 (12.913)
Realization Status	0.667*** (0.058)				94.864*** (8.750)			
Observations	6,166	4,422	2,329	2,093	6,166	4,422	2,329	2,093
Adjusted R^2	0.056	0.033	0.020	0.069	0.049	0.039	0.029	0.057
Investment Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table VI
Skill and Scalability in PE

This table shows the distribution of the skill and scalability parameters (see table 2 in Barras, Gagliardini, and Scaillet (2022)). The parameters are based on manager-specific OLS regressions of gross PME returns on the natural logarithm of investment size and the natural logarithm of the manager's deal sequence. The sample is restricted to 412 managers with at least five realized deals. Negative scale coefficients indicate diseconomies of scale.

	Moments				Proportions		Quantiles		
	Mean	SD	Skewness	Kurtosis	Negative	Positive	5%	50%	95%
Skill Coefficient	3.69	4.83	1.12	7.95	0.21	0.79	-3.34	3.29	11.07
Scale Coefficient	-0.44	0.97	-1.94	27.29	0.74	0.26	-2.20	-0.54	0.44

- For NPV, the persistence coefficient rises from 0.121 in the early period to 0.196 in the later period.
- The coefficient on log investment size is sharply negative in the PME regressions, confirming that scaling up a deal reduces relative returns.

5.5 Table VI: skill and scalability

Read:

- The average manager in the skill/scalability sample has a skill coefficient of 3.69 and a scale coefficient of -0.44 .
- **79%** of managers have a positive skill coefficient.
- **74%** have a negative scale coefficient, meaning relative returns usually fall as the manager is assigned larger deals.
- A doubling of investment size lowers PME by about $0.44 \times \log(2) \approx 0.30$ for the average manager.

Interpretation. Most managers appear value-creating on small deals, but most also face diseconomies of scale. That is exactly why firms must actively decide who should handle the biggest transactions.

5.6 Table VII and Table VIII: the internal labor market

Table VII

Manager-Performance Sensitivity: Univariate Analysis

This table examines capital reallocation decisions and managerial labor market outcomes based on past deal net present value (NPV, using regional MSCI indices plus 200 bps p.a. as discount rate). The sample is restricted to 4,854 realized deal involvements for which we observe at least one previous realized investment of the same manager with the same PE organization. To determine a manager's prior performance, we benchmark each of the manager's past investments against the median deal NPV in the corresponding PE firm-year (using all of the realized PE firm investments in the deal's investment year). Column (1) shows univariate statistics for the full sample. Column (2) shows deal involvements for managers who outperformed in terms of NPV with at least 50% of their previous deals, while column (3) shows deal involvements for managers with a success rate of previous NPV performances below 50%. Column (4) shows the differences between the out- and underperformers. Column (5) displays the corresponding test statistics. Panel A reports the median for unweighted averages for gross PME and NPV of all previous realized deals. Panel B compares capital reallocation measures to previous deal performance. *Investment Size* is the median size of the current deal. *Investment Size per Manager* is the median size of the current deal divided by the number of managers on the deal. Panel C looks at managerial labor market outcomes within three years after the preceding deal's investment year. *Partner Promotion* is a dummy variable set to one for managers who are promoted to partner, and zero otherwise. For this variable, we restrict the sample to 2,292 deal involvements that occurred when the manager was not yet partner. *Leaves PE Firm (Industry)* is a dummy variable set to one if a manager leaves the current firm (PE industry), and zero otherwise. For these variables, we restrict the sample to the remaining 2,562 partner deal involvements. All tests in Panels A and B are rank-sum tests. Tests in Panel C are chi-square tests. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	Track Record (NPV Success Rate)			(2) – (3)	
	All (1)	Outperformer (2)	Underperformer (3)	Diff (4)	Test Statistic (5)
Obs.	4,854	2,625	2,229		
Panel A: Average Past Performance					
PME	2.05	2.44	1.73	0.71	***
NPV	22.9	44.2	7.5	36.7	***
Panel B: Capital Allocation (Next Deal)					
Inv. Size	38.4	44.6	33.0	11.6	***
Inv. Size per Manager	17.0	18.6	15.4	3.2	***
Panel C: Career Development (Three Years after Deal)					
Partner Promotion	13.9%	15.7%	11.7%	4.0%	***
Leaves PE Firm	2.5%	2.2%	2.9%	-0.7%	
Leaves PE Industry	2.1%	1.5%	2.9%	-1.4%	**

Table VIII

Manager-Performance Sensitivity: Multivariate Analysis

This table examines capital reallocation decisions (model (1)) and managerial labor market outcomes (models (2) to (4)) based on past deal net present value (NPV, using regional MSCI indices plus 200 bps p.a. as discount rate). To measure a manager's track record at any given time, we calculate *NPV Success Rate*, the percentage of past realized deals that were outperforming in terms of NPV. We consider a deal to be outperforming when it delivered a NPV greater than the median of all realized deals in the same investment year by the same PE firm. The sample is restricted to 4,854 deal involvements for which we observe two subsequent deals of the same manager within the same PE organization. In model (1), the dependent variable is the natural logarithm of current investment size divided by the number of managers on the deal. In model (2), the dependent variable is a dummy variable set to one for managers promoted to partner within three years after the preceding deal's investment year, and zero otherwise. For this variable, we restrict the sample to 2,292 deal involvements that occurred when the manager had not been a partner yet. In model (3), the dependent variable is a dummy variable set to one for managers who leave the current firm within three years after the preceding deal's investment year, and zero otherwise. In model (4), the dependent variable is a dummy variable set to one for managers who leave the PE industry within three years after the preceding deal's investment year, and zero otherwise. For the last two variables, we restrict the sample to the remaining 2,562 partner deal involvements. All models are OLS regressions. Standard errors are in parentheses and clustered at the manager level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Dependent variable	Capital Allocation (Next Deal)	Career Development (Three Years after Deal)		
	Log Inv. Size per Manager (1)	Partner (2)	Leaves PE Firm (3)	Leaves PE Industry (4)
NPV Success Rate	0.202*** (0.072)	0.056** (0.026)	-0.020* (0.012)	-0.020*** (0.010)
Log Age at Deal	0.246* (0.148)	-0.045 (0.047)	-0.011 (0.016)	0.040** (0.020)
Female	-0.183 (0.147)	-0.007 (0.055)	-0.024*** (0.006)	-0.021*** (0.004)
MBA Degree	-0.008 (0.075)	-0.053** (0.023)	0.019** (0.009)	0.001 (0.009)
Ivy League	0.239*** (0.080)	0.014 (0.026)	-0.010 (0.011)	0.012 (0.009)
Attendance	-0.071 (0.052)	-0.031* (0.017)	-0.011** (0.006)	-0.006 (0.006)
Log Manager Deal	4,854	2,292	2,562	2,562
Sequence	Adj. R ²	0.067	0.012	0.019
Observations	Yes	Yes	Yes	Yes
Investment Year FE	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes

Read:

- Table VII compares outperformers and underperformers using the NPV success-rate track-record measure.
- Outperformers' next median deal is \$44.6m versus \$33.0m for underperformers.
- Measured per manager, the next deal size is \$18.6m for outperformers versus \$15.4m for underperformers.
- Promotion differences are meaningful: 15.7% of outperformers are promoted to partner within three years, versus 11.7% of underperformers.
- Leaving the PE industry is also more common among underperformers: 2.9% versus 1.5%.
- Table VIII confirms these patterns in multivariate regressions. A unit increase in NPV success rate raises log investment size per manager by 0.202, raises the probability of partner promotion by 5.6 percentage points, and lowers exit probabilities by about 2 percentage points.

Interpretation. The PE organization does not merely observe skill; it appears to reward it both on the intensive margin (larger deals) and the extensive margin (promotion and retention).

5.7 Table IX: how much value comes from the organization itself?

Read:

- Table IX asks whether managers generate more NPV when the PE firm significantly increases the capital assigned to them.

Table IX
Capital Reallocations and NPV

This table reports results of OLS regressions of current *NPV per Manager* on a PE firm's capital allocation decisions. The analysis is at the manager-deal involvement level and includes realized deals only. The dependent variable is the current deal's NPV divided by the number of managers on the same deal. *Investment Size per Manager Increase* is a dummy variable equal to one if a manager's current deal investment size per manager (investment size divided by the number of managers on the deal) exceeds their previous deal's investment size per manager by a specified threshold, and zero otherwise. *Investment Size per Manager Decrease* is a dummy variable equal to one if a manager's current investment size per manager falls below their previous investment size per manager by the same threshold, and zero otherwise. Models (1) to (3) use a threshold of 50%. Model (2) focuses on capital allocations within the same fund. Model (3) uses PE firm instead of fund fixed effects. Models (4) to (6) use alternative thresholds for the dependent variable, with thresholds of 20%, 30%, and 40%, respectively. Standard errors are in parentheses and clustered at the manager level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Dep. Variable	NPV per Manager					
	Threshold 50/150			Alternative Thresholds (Baseline)		
	Baseline	Same Fund	PE Firm FE	Threshold 80/120	Threshold 70/130	Threshold 60/140
Test	(1)	(2)	(3)	(4)	(5)	(6)
Inv. Size per Manager Increase	12.883** (5.645)	16.454* (9.473)	12.679*** (4.857)	11.001* (6.233)	11.687** (5.401)	13.902** (5.554)
Inv. Size per Manager Decrease	6.131 (7.439)	8.786 (11.208)	3.727 (6.538)	6.391 (6.698)	7.116 (6.639)	8.689 (6.766)
Log Inv. Size per Manager	13.096*** (5.036)	20.344** (9.627)	17.600*** (5.109)	13.829*** (4.675)	13.876*** (4.835)	13.672*** (4.962)
Log Holding Period	-3.654 (5.083)	-4.248 (7.477)	-3.714 (4.733)	-3.964 (5.076)	-3.625 (5.104)	-3.657 (5.077)
Log Manager Deal Sequence	10.496 (9.850)	9.371 (16.637)	-11.101 (10.091)	10.615 (9.886)	10.464 (9.898)	10.326 (9.842)
Observations	3,611	2,199	3,611	3,611	3,611	3,611
Adjusted R ²	0.167	-0.398	-0.006	0.166	0.166	0.167
Average NPV (m\$)	31.7	27.1	31.7	31.7	31.7	31.7
Investment Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Manager FE	Yes	Yes	Yes	Yes	Yes	Yes
Fund FE	Yes	Yes	No	Yes	Yes	Yes
PE Firm FE	No	No	Yes	No	No	No

- In the baseline regression, a large increase in investment size per manager is associated with an extra \$12.9m of NPV per manager.
- Since average NPV in that subsample is \$31.7m, this implies that about **41%** of NPV is associated with capital-allocation decisions by the PE firm.
- In the cleaner within-fund version, the coefficient rises to \$16.5m against an average NPV of \$27.1m, implying a contribution of roughly **61%**.
- Across alternative thresholds, the implied share typically lies in the **33%–45%** range.

Interpretation. The organization matters not just because it employs skilled people, but because it reallocates scarce capital toward the people who can turn that capital into the most NPV.

6 Short summary

- **Method.** The paper combines proprietary buyout cash-flow data with hand-collected manager attribution and career data to study persistence, skill, scalability, internal capital allocation, and organizational value creation in PE.
- **Main conceptual result.** The correct measure of PE managerial skill when funds scale up is NPV, not just relative-return measures such as PME.
- **Main empirical result.** Fund-level and manager-level PME persistence weaken over time, but NPV persistence strengthens. Most managers exhibit positive skill and negative scalability.
- **Internal labor market result.** Managers with stronger NPV track records receive larger subsequent deals, are more likely to be promoted, and are less likely to leave the industry.

- **Organizational result.** Roughly 40% of PE NPV is associated with the PE firm’s internal capital allocation decisions.
- **Takeaway.** In PE, declining persistence in relative returns should not be interpreted as declining skill. Instead, it may reflect organizations scaling up their best managers and thereby transforming skill into dollar value-added rather than into persistent PME outperformance.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that the growth of PE has changed the meaning of performance persistence.
- Relative returns become less persistent as deals and funds become larger, but value creation measured by NPV remains persistent and even becomes more so.
- The paper also establishes that PE firms allocate more capital to managers who have stronger NPV track records and that these internal allocations account for a substantial fraction of value creation.

7.2 Economic intuition

- Successful PE firms want to raise bigger funds because fees and carry do not increase much on a percentage basis with success.
- Bigger funds generally imply bigger deals, and bigger deals are harder to earn very high *relative* returns on.
- So success changes the scale at which managers operate. A manager who is skillful enough to be promoted to bigger deals may no longer look exceptional on PME, even while creating more dollars of value.
- The PE organization’s role is to learn which managers can best handle this scale-up problem and then to allocate capital accordingly.

7.3 Takeaway

This paper is important because it links the modern “skill versus scale” logic from mutual funds to private equity and then shows that the organization itself is part of the value-creation technology. In PE, the right question is not only *who earned the highest relative return?* It is also *who can be trusted with more capital and still generate the most NPV?* The paper’s answer is that PE firms appear to know a meaningful amount about this, and that their internal capital allocation decisions are a first-order source of value.